FUNCTIONAL SPECIFICATION

Skill Exchange Platform Blueprint

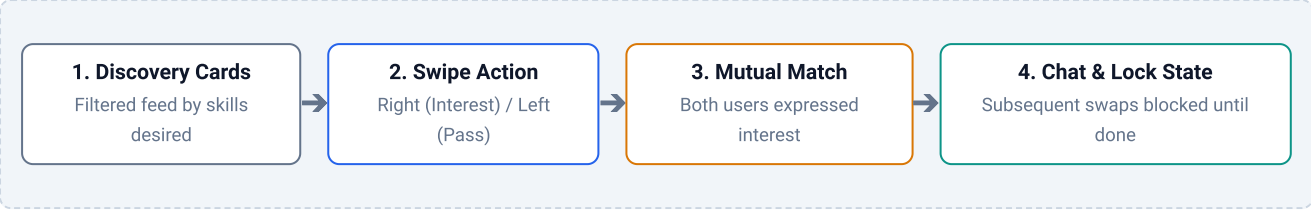
Swipe-Based Skill Barter, Mutual Matching & Gamified Learning Ecosystem

Structure: 2 Builders
Frontend: Client Lead
Backend: Server Lead

1. CORE PRODUCT VISION

A peer-to-peer skill barter marketplace where users swipe on profiles based on skills offered versus skills desired. Learning is structured around dedicated 1:1 barter sessions with lock mechanics that keep users focused until learning goals are achieved.

2. SWIPE ENGINE & EXCHANGE LOCK STATE MACHINE



3. SWIPE MECHANICS & SUBSCRIPTION TIERS

Swipe Right / Left Interaction

Users review candidate cards displaying Skills Provided, Skills Wanted, bio, and rating. Swipe Right indicates interest in bartering; Swipe Left skips the candidate.

Active Skill Exchange Lock

Once a mutual match occurs, a 1:1 Chat unlocks. To prevent overload, **all other swiping and new exchanges are locked** until both parties confirm learning completion.

Free Tier Quota

10 Swipes / Day (Reset at 00:00 UTC)

Paid / Premium Tier

50 Swipes / Day + Priority Feed Placement

4. PROFILE SYSTEM & RECOMMENDATION ENGINE

User Identity & Matching Profile

- **Personal Information:** Full Name, Profile Picture, Bio, and Email Address.
- **Security & Verification:** Phone Number binding with SMS OTP (One-Time Password) verification to ensure authentic, non-spam profiles.
- **Skills Taxonomy:** Explicit listing of *Skills Provided* (what I teach) and *Skills Wanted* (what I want to learn).
- **Matching Recommendation Algorithm:** The home feed algorithm queries profiles where Candidate's `Skills Provided` intersects with User's `Skills Wanted` and vice-versa for reciprocal compatibility.

5. LEARNING BLOG & STREAK GAMIFICATION ENGINE

Public Learning Logs & Reward System

To foster active engagement and document knowledge sharing, users can write micro-blogs detailing key takeaways from their barter sessions.

- **Daily Post Streak:** Writing at least one blog post or learning update every 24 hours increments the user's active streak counter.
- **Streak Bonus Swipes:** Maintaining consecutive day streaks rewards users with **extra bonus swipes** on top of their daily free quota (e.g., +5 bonus swipes for a 3-day streak).
- **Public Proof of Skill:** Blogs are displayed on user profiles as proof of teaching ability and active community participation.

6. ACCOUNT LIFECYCLE & PRIVACY CONTROLS

Settings & Data Management Flow

- **Profile Customization:** Dynamic updating and altering of bio, skills list, and availability status.
- **Notification Preferences:** Toggle controls for new match alerts, direct messages, and streak reminders.
- **Permanent Account Deletion:** Full account purge mechanism compliant with data privacy laws. Deleting an account removes profile data, active match queues, blog posts, and invalidates active authentication tokens.

7. DETAILED TEAM TASK ALLOCATION MATRIX

Specific task breakdown tailored to Frontend/UI/Client Side and Backend/Server Side responsibilities.

Frontend Lead (Client / UI / UX)

Primary Focus: Interactive UI, Swipe Gestures, State Management.

- **Swipe Deck Component:** Tinder-style swipe card stack with touch/drag gestures and animation effects.
- **OTP Verification UI:** Phone input screen with auto-advancing 6-digit PIN input fields.
- **Match & Lock Screen:** Overlay notification upon mutual match and locked view disabling search during active exchanges.
- **1:1 Chat Workspace:** Real-time messaging layout with "Mark Learning Complete" agreement modal.
- **Blog Editor & Streak Widget:** Rich text post composer and streak counter badge in header.
- **Settings Dashboard:** Forms for account edits, notification toggles, and delete account confirmation modal.

Backend Lead (Server / DB / API)

Primary Focus: Logic Engine, Security, Data Models, Real-Time.

- **Auth & OTP Service:** Integration with SMS gateway (Twilio) and JWT session generation.
- **Swipe & Match Engine:** API for logging right/left swipes, checking mutual interest, and swipe count reset crons.
- **Exchange Lock Manager:** Middleware checking active exchange status before allowing feed access.
- **Blog & Streak Cron:** CRUD APIs for blogs and daily scheduled worker calculating active user streaks & bonus credits.
- **WebSocket Chat Server:** Socket.io message relay with persistence in DB.
- **Account Lifecycle API:** Cascade deletion endpoints removing user records and media assets.

8. MILESTONES & EXECUTION ROADMAP

Sprint	Milestone	Frontend Tasks	Backend Tasks
Sprint 1	Auth & Profile	Signup/Login screens, Phone OTP UI, Profile form.	Twilio OTP integration, User DB schema, Auth tokens.
Sprint 2	Swipe & Match	Swipe deck UI, Mutual match popup, Feed filters.	Swipe API, Daily limit counter, Matching feed query.
Sprint 3	Chat & Lock	Chat interface, "Complete Learning" agreement modal.	Socket.io real-time chat, Active exchange lock engine.
Sprint 4	Blog & Streak	Blog feed, Editor, Streak badge & bonus tracker.	Blog CRUD, Streak cron job, Bonus swipe allocator.
Sprint 5	Settings & Launch	Settings view, Account deletion modal, QA tests.	Data deletion cascade API, Security audit, Deployment.

9. TEAM COLLABORATION & RECOMMENDED FREE TECH STACK

Frontend Tools

- **React / Next.js** Framework.
- **Tailwind CSS** Styling.
- **Framer Motion** Swipe gesture cards.
- **Vercel** Free client hosting.

Backend Tools

- **Node / Express** API server.
- **Supabase** PostgreSQL & Auth.
- **Twilio / Firebase** Phone OTP.
- **Render** Free server hosting.